

# Mathematical Analysis Lecture 17. 微分学

例: 求  $\lim_{x \rightarrow 0} \frac{\cos x - e^{-\frac{x^2}{2}}}{x^4}$  (可用 L'Hospital, 也可用 Taylor 公式:)

$$\lim_{x \rightarrow 0} \frac{\cos x - e^{-\frac{x^2}{2}}}{x^4} = \frac{[1 - \frac{1}{2}x^2 + \frac{1}{4}x^4 + o(x^4)]}{x^4} - \frac{[1 - \frac{1}{2}x^2 - \frac{1}{8}x^4 + o(x^4)]}{x^4} = -\frac{1}{12}, \text{ 展开到同阶}$$

用 Taylor 公式可以解决相当多的极限问题.

例: 求  $\lim_{x \rightarrow 0} \frac{\ln(1+\sin^2 x) - 6(\sqrt[3]{2-\cos x} - 1)}{x^4}$

$$\ln(1+\sin^2 x) = \sin^2 x - \frac{\sin^4 x}{2} + o(\sin^4 x) = [x - \frac{x^3}{6} + o(x^4)]^2 - \frac{1}{2}(x + o(x))^4 = x^2 - \frac{5}{6}x^4 + o(x^4)$$

$$\sqrt[3]{1+(1-\cos x)} = 1 + \frac{1}{3}(1-\cos x) - \frac{1}{9}(1-\cos x)^2 + o((1-\cos x)^2) = 1 + \frac{1}{3}(\frac{x^2}{2} - \frac{x^4}{24} + o(x^4)) - \frac{1}{36}x^4 = 1 + \frac{1}{6}x^2 - \frac{x^4}{24} + o(x^4)$$

$\therefore$  原式 =  $-\frac{7}{12}$  解毕. 关键:  $(1+x)^\alpha$  的麦克劳林公式要求  $x \rightarrow 0$

例: 证: 已知  $f(x)$  在  $[0, 1]$  上有二阶导数,  $|f(x)| \leq A$ ,  $|f''(x)| \leq B$ . 证明:  $|f'(x)| \leq 2A + \frac{1}{2}B$

$f(x) = f(c) + f'(c)(x-c) + \frac{1}{2}f''(\xi)(x-c)^2$  (带 Lagrange 余项的 Taylor series)  $c \in [0, 1], x \in [0, 1], \xi \in [0, 1]$

$$f(0) = f(c) + f'(c)(0-c) + \frac{1}{2}f''(\xi_1)c^2, \quad f(1) = f(c) + f'(c)(1-c) + \frac{1}{2}f''(\xi_2)(1-c)^2$$

$$\therefore f'(c) = \frac{f(1) - f(0)}{1-0} + \frac{1}{2}[f''(\xi_2)(1-c)^2 - f''(\xi_1)c^2] \leq 2A + \frac{B}{2}$$

例: 设  $f(x)$  在  $[a, b]$  上二阶可导,  $f(a) = f(b) = 0$ , 证明  $\max_{a \leq x \leq b} |f(x)| \leq \frac{1}{8}(b-a)^2 \max_{a \leq x \leq b} |f''(x)|$

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2}f''(\xi_1)(x-x_0)^2 \quad \text{for } x=a, b$$

$$\therefore 0 = f(x_0) + \frac{1}{2}f''(\xi_1)(a-x_0)^2 = f(x_0) + \frac{1}{2}f''(\xi_2)(b-x_0)^2$$

$$\therefore |f(x_0)| \leq \frac{1}{2}(a-x_0)^2 |f''(x)|_{\max} \quad |f(x_0)| \leq \frac{1}{2}(b-x_0)^2 |f''(x)|_{\max}$$

$$(\frac{a-b}{2})^2 \geq \max\{(a-x_0)^2, (b-x_0)^2\} \quad \therefore |f(x_0)| \leq \frac{1}{8}(a-b)^2 |f''(x_0)|_{\max}. \text{ 结果导向, 构造系数}$$

在特殊点展开为佳, 然后用其他点代入消一肩为好.

求曲线的渐近线方程. 定义: 若  $\exists y = ax + b$ , 对点  $(x, f(x))$ , 当  $x \rightarrow +\infty$  or  $x \rightarrow -\infty$ ,  $\lim d = 0$ , 则  $y = ax + b$  是一条渐近线.

$$\text{即 } \lim_{x \rightarrow +\infty} [f(x) - (ax+b)] = 0 \text{ 或 } \lim_{x \rightarrow -\infty} [f(x) - (ax+b)] = 0$$

$$\text{等价于 } a = \lim_{x \rightarrow +\infty} \frac{f(x)}{x} \text{ 或 } a = \lim_{x \rightarrow -\infty} \frac{f(x)}{x}. \text{ 然后 } b = \dots \text{ (待 } a \text{ 求)}$$

若  $x \rightarrow a^+$  or  $a^-$  时,  $f(x) \rightarrow +\infty$  or  $-\infty$ , 则  $x=a$  是  $f(x)$  的一条垂直渐近线.

e.g 求  $y = x^3 [e^{\frac{1}{x}} + e^{-\frac{1}{x}} - 2]$  的渐近线

$$a = \lim_{x \rightarrow \infty} \frac{y}{x} = \lim_{x \rightarrow \infty} (x^2 (2 + \frac{1}{x} + \frac{1}{2x^2} + o(\frac{1}{x^2}) - \frac{1}{x} + \frac{1}{2x^2} + o(\frac{1}{x^2}) - 2)) = 1.$$

求铅直渐近线! 万一存在呢?